Bluetooth Controlled DATA LOGGER ROBOT For Soil Testing

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This robot can be controlled with a smartphone using Bluetooth through an Android app. Its three sensors can measure four parameters: temperature, humidity, soil moisture, and ambient light intensity in greenhouses, farms, gardens, parks, etc.

Usually robots like robotic hand, agriculture robot, fire-fighting robot, spy robot, snake robot, humanoid, bomb (or mine) diffusing robot operate automatically without any human intervention or are remote-controlled. Remote-controlled robots are mostly wireless.

There are different types of wireless remote-controlled robots like:

1. Fire-fighting robots, which are used as fire extinguishers for spraying water or cardon-dioxide on fire.

2. Bomb (mine) diffusing robots, which are used to diffuse live bombs or mines in a battlefield.

3. Snake robots, which can enter small tunnels or pipelines for search and rescue operations or to find out any problem like leakage in pipelines

This robot monitors the ambient parameters in a field. The operator can manoeuvre the robot in a radius of 10 to 30 metres and take measurements to select the best place for plantation. And if plantation is already done, the above-mentioned four parameters can be checked to see whether these are within the threshold levels or not for taking any corrective actions.

This project can be modified for some other applications also by just changing the sensors. For example, by equipping the robot with MQ2, MQ3 or similar gas sensors, it can be used to detect leakage of any gas like cardon-dioxide, methane, or LPG.

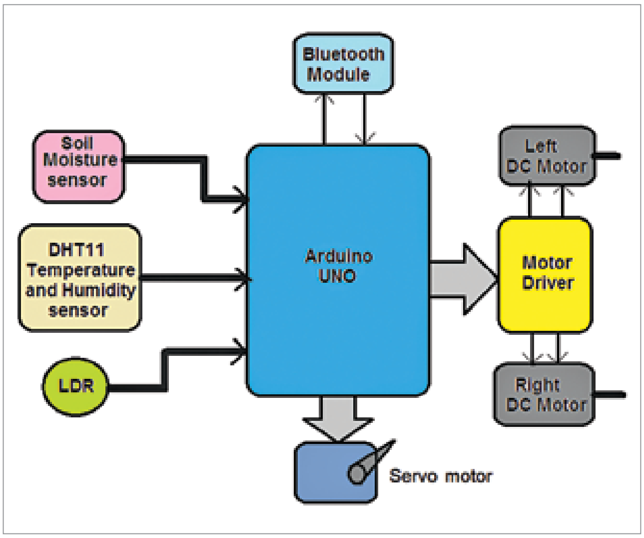

Fig. 1: Block diagram of Bluetooth-controlled data logger robot

Working

Before building the robot, let us first understand its working through the system block diagram shown in Fig. 1 and the circuit diagram in Fig. 2. The major building blocks of the system include the three sensors (soil moisture, DHT11 temperature and humidity sensor, and LDR), an Arduino Uno development board, Bluetooth module HC-05, DC servo motor, two DC gear motors, and motor driver chip L293D. Let us first understand the role of the major components used in the project.

Soil moisture sensor (SM1) is used to measure moisture content in soil and give analogue voltage output as per the moisture level. Its output voltage decreases as moisture content increases.

DHT11 temperature and humidity sensor (SM2) is a smart sensor that measures atmospheric temperature and humidity and gives direct digital values for temperature in °C and for humidity in % RH.

The light dependant resistor (LDR1) is a photo-conductive device whose resistance decreases as light intensity increases, and vice versa.

Bluetooth module (HC-05) is used to command (to move forward, reverse, left, and right) the robot from a smartphone with the help of Arduino Uno microcontroller.

The Arduino Uno board (Board1) performs following tasks:

- Reads analogue output voltage from soil moisture sensor and converts it into digital value. Then it calibrates it between 0 and 100% moisture level
- Reads temperature and humidity values from DHT11 sensor
- Reads analogue voltage output from LDR1 and calibrates light intensity between 0 and 100%
- Gets different commands from Bluetooth module and rotates two DC motors to move the robot forward, reverse, left, right, or stop it.
- Sends (transmits) readings of